Deal memo: Henry Hub gas storage, Nov 2026 to Oct 2027

Status: DRAFT for discussion. Valuation date 2026-10-14. Gas prices are real EIA Henry Hub; the structure is a hypothetical term sheet, not a real counterparty.

Recommendation

Value the deal at USD 164,937 (least-squares Monte Carlo, standard error 2,968), of which USD 34,667 is intrinsic and USD 130,270 is extrinsic. Lock the intrinsic with forwards, re-optimise monthly (rolling intrinsic, mean USD 165,122) and quote the extrinsic only at a discount: it exists only if the desk actually re-hedges. The minimum fee for a 12% return on capital is USD -81,113; a negative figure means the deal clears the hurdle with no fee under these assumptions.

Term sheet

Item	Value
Capacity	1,000,000 MMBtu
Injection / withdrawal rate	200,000 / 300,000 MMBtu per month
Injection / withdrawal cost	0.02 / 0.02 USD per MMBtu
Fuel loss (in kind)	1%
Start / end inventory	0 / 0
Discount rate	4%

Valuation and cross-checks

Method	Value USD	Note
Intrinsic (linear programme, forward curve)	34,667	exact for the given curve
LSMC, no jumps	164,937	SE 2,968; lower bound
Lattice dynamic programme	169,508	within noise of LSMC
LSMC with price jumps	535,556	jump risk raises option value
LSMC, volatility x1.5	305,165	value rises with volatility

Method check: with volatility set near zero, the linear programme, LSMC and lattice all give USD 33,795.

Risk and economics

Measure	USD
Expected P&L after hedge cost	222,012
Worst 5% outcome	-103,419
VaR 95% vs mean	325,431
ES 99% (capital proxy)	633,412
Return on capital	35% against a 12% hurdle

Hedging with rolling intrinsic cut the 99% shortfall from USD 649,050 to USD 633,412: a modest reduction, because the extrinsic value is itself the exposure to price moves.

Limits and assumptions

- One-factor model with jumps calibrated to Henry Hub history since 2008; storage is priced on a monthly grid.
- Capital is an Expected Shortfall proxy and the 12% hurdle and USD 0.01 bid-offer are stated assumptions, not any firm's policy.
- Counterparty credit, operational constraints and seasonal injection limits beyond the monthly rates are not modelled.
- Henry Hub is a US price; it stands in for the structure and is not a European gas hub.